

LECTURE NOTES

Hard Problems in Cryptography

Mathematical Foundations for Post-Quantum Cryptography

Factorization · Discrete Logarithm · Lattices · Codes · Isogenies · Multivariate · Hash

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Classical hardness assumptions and post-quantum reductions

Hard Problems in Cryptography

Mathematical Foundations for Classical and Post-Quantum Cryptography

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Chapter 1

Preliminaries

1.1 Search vs. decision vs. distinguishing.

In post-quantum cryptography (PQC), many security assumptions are stated using one of three of hard problems:

1. **Search problem:** recover the hidden object.
2. **Decision problem:** answer a yes/no question about the instance.
3. **Distinguishing problem:** tell whether samples come from one distribution or another.

These three look similar, but they play different roles in theory and in cryptographic reductions.

Type	**Lattice**	**Code**
Search	SVP, CVP, BDD, SIS, Search-LWE	Syndrome Decoding, decoding, low-weight codeword search
Decision	GapSVP, GapCVP	existence of error of weight $\leq w$, gap decoding forms
Distinguishing	Decision-LWE, Ring-LWE, Module-LWE	structured code vs. random code, disguised secret code vs. random linear code

1.2 Preliminaries for Lattice-Based Hard Problems

This section collects the mathematical background needed to understand lattices, lattice basis reduction, and the hard problems that arise in post-quantum cryptography.

1.2.1 Notation

$\mathbb{Z}$ the ring of integers.

$\mathbb{R}$ the field of real numbers.

$\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$ integers modulo q.

$\mathbb{R}^n$ the n-dimensional real vector space.

$\mathbf{x}, \mathbf{y}, \mathbf{b}$ vectors.

A, B matrices.

$\langle \mathbf{x}, \mathbf{y} \rangle$ the standard inner product.

$\|\mathbf{x}\|$ the Euclidean norm of $\mathbf{x}$.

$\det(B)$ determinant of a square matrix B.

$\mathcal{L}(B)$ the lattice generated by the columns of B.

1.2.2 Vector spaces and Euclidean geometry

Vectors in $\mathbb{R}^n$

Definition 1.2.1. A vector in $\mathbb{R}^n$ is an ordered n-tuple

$$\mathbf{x} = (x_1, x_2, \ldots, x_n), \qquad x_i \in \mathbb{R}.$$

The set of all such vectors forms a vector space over $\mathbb{R}$.

Inner product

Definition 1.2.2. The standard inner product of two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n.$$

This quantity measures the geometric alignment of two vectors.

Remark 1.2.3. Two vectors are said to be *orthogonal* if

$$\langle \mathbf{x}, \mathbf{y} \rangle = 0.$$

Euclidean norm

Definition 1.2.4. The Euclidean norm of a vector $\mathbf{x} \in \mathbb{R}^n$ is

$$\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

Remark 1.2.5. This norm satisfies the standard properties:

1. $\|\mathbf{x}\| \geq 0$, with equality if and only if $\mathbf{x} = \mathbf{0}$,
2. $\|\alpha \mathbf{x}\| = |\alpha| \, \|\mathbf{x}\|$ for $\alpha \in \mathbb{R}$,
3. $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ (triangle inequality).

Linear Independence

Definition 1.2.6. A finite set of vectors

$$\{\mathbf{b}_1, \ldots, \mathbf{b}_m\} \subseteq \mathbb{R}^n$$

is **linearly independent** if

$$a_1 \mathbf{b}_1 + \cdots + a_m \mathbf{b}_m = \mathbf{0}$$

implies

$$a_1 = \cdots = a_m = 0.$$

Remark 1.2.7 (span)**.** The real span of $\mathbf{b}_1, \ldots, \mathbf{b}_m$ is

$$V = \operatorname{span}_{\mathbb{R}}(\mathbf{b}_1, \ldots, \mathbf{b}_m) = \left\{ \sum_{i=1}^{m} a_i \mathbf{b}_i : a_i \in \mathbb{R} \right\}.$$

So V is just the set of all vectors you can build from $\mathbf{b}_1, \ldots, \mathbf{b}_m$ using real coefficients.

Remark 1.2.8. Let B be the matrix whose columns are $\mathbf{b}_1, \ldots, \mathbf{b}_m$:

$$B = \begin{bmatrix} | & & | \\ \mathbf{b}_1 & \cdots & \mathbf{b}_m \\ | & & | \end{bmatrix}.$$

Then

$$B\mathbf{a} = \begin{bmatrix} | & & | \\ \mathbf{b}_1 & \cdots & \mathbf{b}_m \\ | & & | \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix} = a_1 \begin{bmatrix} | \\ \mathbf{b}_1 \\ | \end{bmatrix} + \cdots + a_m \begin{bmatrix} | \\ \mathbf{b}_m \\ | \end{bmatrix} = a_1 \mathbf{b}_1 + \cdots + a_m \mathbf{b}_m.,$$

So

$$V = \operatorname{span}_{\mathbb{R}}\{\mathbf{b}_1, \ldots, \mathbf{b}_m\} = \left\{ \sum_{i=1}^{m} a_i \mathbf{b}_i : a_i \in \mathbb{R} \right\} = \{B\mathbf{a} : \mathbf{a} \in \mathbb{R}^m\}.$$

The first writes the span as a sum. The second writes the same thing using matrix multiplication.

Lattice

Definition 1.2.9. A lattice $\Lambda \subset \mathbb{R}^n$ is called a lattice of rank m if

$$\Lambda = \left\{ \sum_{i=1}^{m} z_i \mathbf{b}_i : z_i \in \mathbb{Z} \right\},$$

where $\mathbf{b}_1, \ldots, \mathbf{b}_m \in \mathbb{R}^n$ are linearly independent.

Remark 1.2.10.

- If $m = n$, then Λ is called a full-rank lattice.
- If $m < n$, then Λ is a non-full-rank lattice, and it lies in the m-dimensional subspace

$$\operatorname{span}_{\mathbb{R}}\{\mathbf{b}_1, \ldots, \mathbf{b}_m\} \subsetneq \mathbb{R}^n.$$

Remark 1.2.11 (Full-rank Lattice)**.** A **full-rank lattice** $\Lambda \subset \mathbb{R}^n$ is a set of the form

$$\Lambda = \mathcal{L}(B) = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\},$$

where $B \in \mathbb{R}^{n \times n}$ is an invertible matrix. If $B = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \end{bmatrix}$, then equivalently

$$\mathcal{L}(B) = \{z_1\mathbf{b}_1 + \cdots + z_n\mathbf{b}_n : z_1, \ldots, z_n \in \mathbb{Z}\}.$$

Thus, a lattice is the set of all integer linear combinations of n linearly independent vectors in $\mathbb{R}^n$.

1.2.3 Orthogonality and Gram–Schmidt Orthogonalization

1.2.4 Fundamental Parallelepipeds and Determinants

Fundamental Parallelepiped

Let

$$B = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \\ | & | & \cdots & | \end{bmatrix} \in \mathbb{R}^{n \times n}$$

be a basis matrix for a full-rank lattice

$$\Lambda = \mathcal{L}(B) = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\}.$$

Thus the columns $\mathbf{b}_1, \ldots, \mathbf{b}_n \in \mathbb{R}^n$ are linearly independent, and every lattice point is an integer linear combination of these basis vectors.

Fundamental Parallelepiped

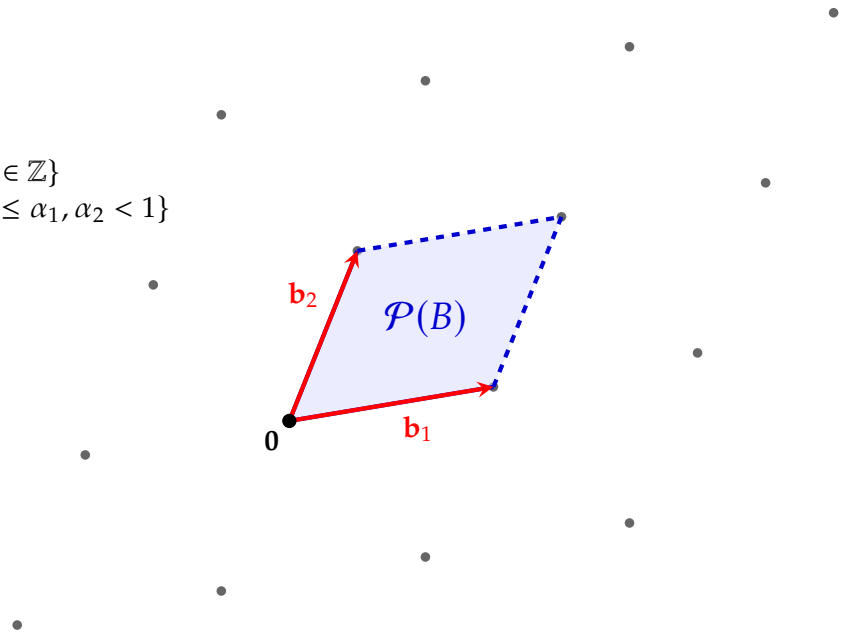
Definition 1.2.12. The **fundamental parallelepiped** associated with B is defined by

$$\mathcal{P}(B) = \left\{ \sum_{i=1}^{n} \alpha_i \mathbf{b}_i : 0 \le \alpha_i < 1 \right\}.$$

Equivalently, in matrix form,

$$\mathcal{P}(B) = \{B\boldsymbol{\alpha} : \boldsymbol{\alpha} \in [0,1)^n\}.$$

$\Lambda = \{z_1\mathbf{b}_1 + z_2\mathbf{b}_2 : z_1, z_2 \in \mathbb{Z}\}$
$\mathcal{P}(B) = \{\alpha_1\mathbf{b}_1 + \alpha_2\mathbf{b}_2 : 0 \le \alpha_1, \alpha_2 < 1\}$



The set $\mathcal{P}(B)$ is the higher-dimensional analogue of:

- an interval in dimension 1,
- a parallelogram in dimension 2,
- a parallelepiped in dimension 3.

Tiling Property

Proposition 1.2.13. *Let $B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_n] \in \mathbb{R}^{n\times n}$ be invertible, and let*

$$\Lambda = \mathcal{L}(B) = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\}$$

be the lattice generated by the columns of B. Define the fundamental parallelepiped

$$\mathcal{P}(B) = \left\{\sum_{i=1}^{n} \alpha_i \mathbf{b}_i : 0 \le \alpha_i < 1\right\} = \{B\boldsymbol{\alpha} : \boldsymbol{\alpha} \in [0,1)^n\}.$$

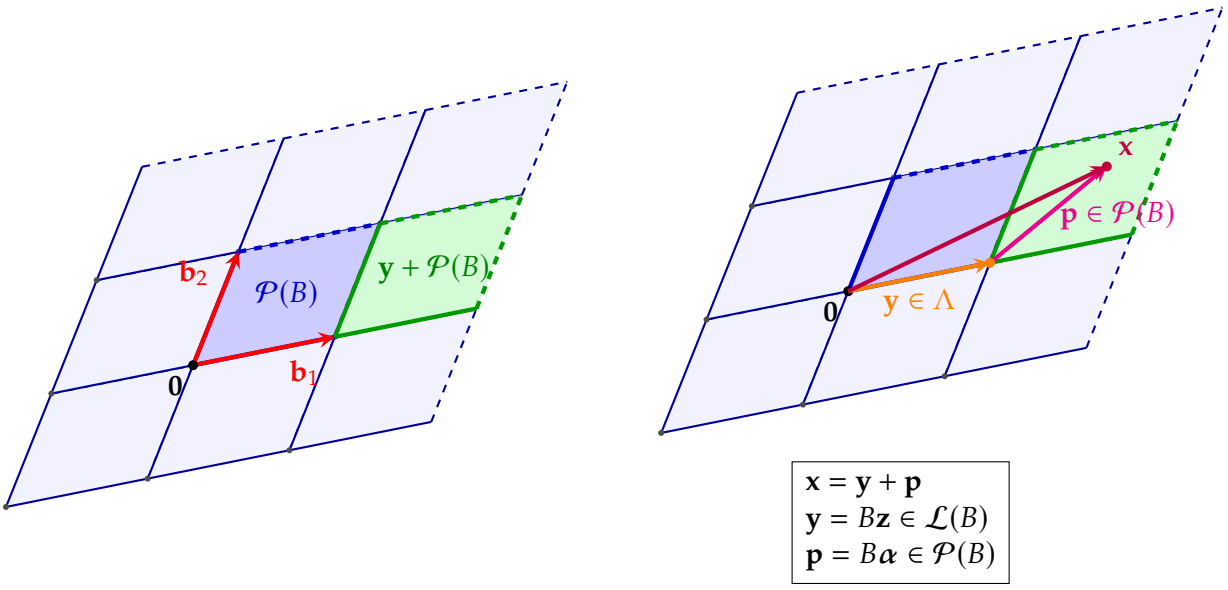
Then:

1. *$\mathbb{R}^n$ is covered by translates of $\mathcal{P}(B)$ by lattice vectors:*

$$\mathbb{R}^n = \bigcup_{\mathbf{y}\in\Lambda} (\mathbf{y} + \mathcal{P}(B)).$$

2. *If $\mathbf{y}_1, \mathbf{y}_2 \in \Lambda$ with $\mathbf{y}_1 \neq \mathbf{y}_2$, then*

$$(\mathbf{y}_1 + \mathcal{P}(B)) \cap (\mathbf{y}_2 + \mathcal{P}(B)) = \varnothing.$$



Proof. Since B is invertible, for every $\mathbf{x} \in \mathbb{R}^n$ there exists a unique vector $\mathbf{t} \in \mathbb{R}^n$ such that

$$\mathbf{x} = B\mathbf{t}.$$

Write

$$\mathbf{t} = \begin{pmatrix} t_1 \\ \vdots \\ t_n \end{pmatrix}.$$

For each i, let

$$z_i = \lfloor t_i \rfloor \in \mathbb{Z}, \qquad \alpha_i = t_i - z_i.$$

Then

$$0 \le \alpha_i < 1 \qquad \text{for each } i,$$

and $t_i = z_i + \alpha_i$. Hence

$$\mathbf{t} = \mathbf{z} + \boldsymbol{\alpha}, \quad \text{where} \quad \mathbf{z} = \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} \in \mathbb{Z}^n, \qquad \boldsymbol{\alpha} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \in [0,1)^n.$$

Multiplying by B, we obtain

$$\mathbf{x} = B\mathbf{t} = B\mathbf{z} + B\boldsymbol{\alpha}.$$

Now $B\mathbf{z} \in \Lambda$, and $B\boldsymbol{\alpha} \in \mathcal{P}(B)$. Therefore

$$\mathbf{x} \in B\mathbf{z} + \mathcal{P}(B) \subseteq \bigcup_{\mathbf{y}\in\Lambda} (\mathbf{y} + \mathcal{P}(B)).$$

Since $\mathbf{x} \in \mathbb{R}^n$ was arbitrary, it follows that

$$\mathbb{R}^n = \bigcup_{\mathbf{y}\in\Lambda} (\mathbf{y} + \mathcal{P}(B)).$$

It remains to prove disjointness. Suppose

$$\mathbf{x} \in (\mathbf{y}_1 + \mathcal{P}(B)) \cap (\mathbf{y}_2 + \mathcal{P}(B))$$

for some $\mathbf{y}_1, \mathbf{y}_2 \in \Lambda$. Then there exist

$$\boldsymbol{\alpha}, \boldsymbol{\beta} \in [0,1)^n$$

such that

$$\mathbf{x} = \mathbf{y}_1 + B\boldsymbol{\alpha} = \mathbf{y}_2 + B\boldsymbol{\beta}.$$

Since $\mathbf{y}_1, \mathbf{y}_2 \in \Lambda$, there exist $\mathbf{z}_1, \mathbf{z}_2 \in \mathbb{Z}^n$ such that

$$\mathbf{y}_1 = B\mathbf{z}_1, \qquad \mathbf{y}_2 = B\mathbf{z}_2.$$

Therefore

$$B\mathbf{z}_1 + B\boldsymbol{\alpha} = B\mathbf{z}_2 + B\boldsymbol{\beta},$$

so

$$B(\mathbf{z}_1 - \mathbf{z}_2) = B(\boldsymbol{\beta} - \boldsymbol{\alpha}).$$

Because B is invertible,

$$\mathbf{z}_1 - \mathbf{z}_2 = \boldsymbol{\beta} - \boldsymbol{\alpha}.$$

The only integer in $(-1, 1)$ is 0, so

$$\boldsymbol{\beta} - \boldsymbol{\alpha} = \mathbf{0}.$$

Hence

$$\boldsymbol{\alpha} = \boldsymbol{\beta} \qquad \text{and} \qquad \mathbf{z}_1 = \mathbf{z}_2.$$

Therefore

$$\mathbf{y}_1 = B\mathbf{z}_1 = B\mathbf{z}_2 = \mathbf{y}_2.$$

This proves that if $\mathbf{y}_1 \neq \mathbf{y}_2$, then

$$(\mathbf{y}_1 + \mathcal{P}(B)) \cap (\mathbf{y}_2 + \mathcal{P}(B)) = \varnothing.$$

□

Uniqueness of Representation Modulo the Lattice

Theorem 1.2.14. *Let $B \in \mathbb{R}^{n\times n}$ be invertible, and let*

$$\Lambda = \mathcal{L}(B) = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\}$$

be the lattice generated by the columns of B. Let

$$\mathcal{P}(B) = \{B\boldsymbol{\alpha} : \boldsymbol{\alpha} \in [0,1)^n\}$$

be the associated fundamental parallelepiped. Then every vector $\mathbf{x} \in \mathbb{R}^n$ can be written uniquely in the form

$$\mathbf{x} = \mathbf{y} + \mathbf{p}, \qquad \mathbf{y} \in \mathcal{L}(B),\ \mathbf{p} \in \mathcal{P}(B).$$

Equivalently, for every $\mathbf{x} \in \mathbb{R}^n$, there exist unique $\mathbf{z} \in \mathbb{Z}^n$, $\boldsymbol{\alpha} \in [0,1)^n$ such that

$$\mathbf{x} = B\mathbf{z} + B\boldsymbol{\alpha} = B(z + \boldsymbol{\alpha}).$$

Proof. Let $\mathbf{x} \in \mathbb{R}^n$ be arbitrary. Since B is invertible, there exists a unique vector

$$\mathbf{t} = \begin{pmatrix} t_1 \\ \vdots \\ t_n \end{pmatrix} \in \mathbb{R}^n$$

such that $\mathbf{x} = B\mathbf{t}$.

(Existence) For each coordinate t_i, define

$$z_i = \lfloor t_i \rfloor \in \mathbb{Z}, \qquad \alpha_i = t_i - z_i.$$

Then

$$0 \le \alpha_i < 1 \qquad \text{for each } i,$$

and $t_i = z_i + \alpha_i$. Hence

$$\mathbf{t} = \mathbf{z} + \boldsymbol{\alpha}, \quad \text{where} \quad \mathbf{z} = \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} \in \mathbb{Z}^n, \qquad \boldsymbol{\alpha} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \in [0,1)^n.$$

Multiplying by B, we obtain

$$\mathbf{x} = B\mathbf{t} = B(\mathbf{z} + \boldsymbol{\alpha}) = B\mathbf{z} + B\boldsymbol{\alpha}.$$

Now $B\mathbf{z} \in \Lambda$ by definition of the lattice, and $B\boldsymbol{\alpha} \in \mathcal{P}(B)$ by definition of the fundamental parallelepiped. Therefore

$$\mathbf{x} = \mathbf{y} + \mathbf{p}$$

with

$$\mathbf{y} = B\mathbf{z} \in \Lambda, \qquad \mathbf{p} = B\boldsymbol{\alpha} \in \mathcal{P}(B).$$

(Uniqueness) Suppose that

$$\mathbf{x} = \mathbf{y}_1 + \mathbf{p}_1 = \mathbf{y}_2 + \mathbf{p}_2,$$

where

$$\mathbf{y}_1, \mathbf{y}_2 \in \Lambda, \qquad \mathbf{p}_1, \mathbf{p}_2 \in \mathcal{P}(B).$$

Since $\mathbf{y}_1, \mathbf{y}_2 \in \Lambda$, there exist $\mathbf{z}_1, \mathbf{z}_2 \in \mathbb{Z}^n$ such that

$$\mathbf{y}_1 = B\mathbf{z}_1, \qquad \mathbf{y}_2 = B\mathbf{z}_2.$$

Since $\mathbf{p}_1, \mathbf{p}_2 \in \mathcal{P}(B)$, there exist $\boldsymbol{\alpha}, \boldsymbol{\beta} \in [0,1)^n$ such that

$$\mathbf{p}_1 = B\boldsymbol{\alpha}, \qquad \mathbf{p}_2 = B\boldsymbol{\beta}.$$

Thus

$$B\mathbf{z}_1 + B\boldsymbol{\alpha} = B\mathbf{z}_2 + B\boldsymbol{\beta}.$$

Rearranging gives

$$B(\mathbf{z}_1 - \mathbf{z}_2) = B(\boldsymbol{\beta} - \boldsymbol{\alpha}).$$

Since B is invertible,

$$\mathbf{z}_1 - \mathbf{z}_2 = \boldsymbol{\beta} - \boldsymbol{\alpha}.$$

The only integer in $(-1, 1)$ is 0, so

$$\boldsymbol{\beta} - \boldsymbol{\alpha} = \mathbf{0}.$$

Hence

$$\boldsymbol{\alpha} = \boldsymbol{\beta} \qquad \text{and} \qquad \mathbf{z}_1 = \mathbf{z}_2.$$

Therefore

$$\mathbf{y}_1 = B\mathbf{z}_1 = B\mathbf{z}_2 = \mathbf{y}_2$$

and

$$\mathbf{p}_1 = B\boldsymbol{\alpha} = B\boldsymbol{\beta} = \mathbf{p}_2.$$

So the representation is unique. □

Determinant of a Full-rank Lattice

Let $\Lambda = \mathcal{L}(B)$ be a full-rank lattice in $\mathbb{R}^n$, where $B \in \mathbb{R}^{n \times n}$ is invertible. The *determinant* of the lattice is defined by

$$\det(\mathcal{L}(B)) = |\det(B)|.$$

Example 1.2.15. If

$$B = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix},$$

then

$$\det(B) = 2, \qquad \det(\mathcal{L}(B)) = 2.$$

Hence the area of the fundamental parallelogram is 2.

Discriminant / volume expression

Invariance under Basis Change

1.2.5 Successive Minima and Shortest Vectors

Chapter 2

Lattice-Based Hard Problems

A (full-rank) lattice $\Lambda \subset \mathbb{R}^n$ is

$$\Lambda = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\}$$

for some basis matrix $B \in \mathbb{R}^{n \times n}$. Classic algorithmic problems are

- **SVP (Shortest Vector Problem):** Find $\mathbf{0} \neq \mathbf{v} \in \Lambda$ minimizing $\|\mathbf{v}\|_2$.
- **GapSVP (Decision/SVP approximation):** Given (Λ, d) decide whether $\lambda_1(\Lambda) \leq d$ or $\lambda_1(\Lambda) > \gamma d$.
- **SIVP (Shortest Independent Vectors):** Find n linearly independent vectors of length $\leq \gamma \cdot \lambda_n(\Lambda)$.

The importance for cryptography: average-case problems (LWE/SIS) reduce from worst-case lattice problems (GapSVP/SIVP) under suitable parameters.

2.1 Learning With Errors (LWE)

TBA

2.2 Short Integer Solution (SIS)

TBA

2.3 NTRU Search Problem

TBA

Chapter 3

Code-Based Hard Problems

3.1 Linear Codes and Syndromes

Let $C \subseteq \mathbb{F}_2^n$ be a linear $[n, k]$ code with parity-check matrix $H \in \mathbb{F}_2^{(n-k)\times n}$. For $e \in \mathbb{F}_2^n$, the *syndrome* is $s = He^\top \in \mathbb{F}_2^{n-k}$. Syndrome decoding asks: given H and s, find a low-weight e with syndrome s.

3.2 Syndrome Decoding (SD)

3.3 QC Syndrome Decoding (QCSD)

Appendix A

Exercises for Lattice

A.1 Euclidean Spaces and Lattices

1. Let
$$\mathbf{x} = (1, -2, 3), \qquad \mathbf{y} = (4, 0, -1).$$
Compute $\langle \mathbf{x}, \mathbf{y} \rangle$, $\|\mathbf{x}\|$, $\|\mathbf{y}\|$, and $\|\mathbf{x} + \mathbf{y}\|$.

2. Determine whether the following sets are lattices in $\mathbb{R}^2$:
   (a) $\mathbb{Z}^2$
   (b) $\{(a, \sqrt{2}a) : a \in \mathbb{Z}\}$
   (c) $\{a(1, 0) + b(1/2, 1) : a, b \in \mathbb{Z}\}$
   (d) $\mathbb{Q}^2$

3. Let
$$\mathbf{b}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \qquad \mathbf{b}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$
Describe explicitly the real span of $\mathbf{b}_1, \mathbf{b}_2$, and the lattice generated by them.

4. Show that if $\mathbf{b}_1, \ldots, \mathbf{b}_m \in \mathbb{R}^n$ are linearly independent, then
$$\Lambda = \left\{ \sum_{i=1}^{m} z_i \mathbf{b}_i : z_i \in \mathbb{Z} \right\}$$
is a discrete subgroup of $\mathbb{R}^n$.

5. Give two distinct bases of the same lattice in $\mathbb{R}^2$.

6. Let $B = [\mathbf{b}_1 \ \mathbf{b}_2]$ with
$$\mathbf{b}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad \mathbf{b}_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$
Write $\mathcal{L}(B)$ explicitly.

7. Show that a lattice may have infinitely many bases.

8. Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice. Prove that Λ is countable.

9. Show that every lattice contains the zero vector and is closed under addition and negation.

10. Let
$$\Lambda = \{(2a + b, b) : a, b \in \mathbb{Z}\}.$$
Find a basis matrix B such that $\Lambda = \mathcal{L}(B)$.

A.2 Gram–Schmidt Orthogonalization

A.3 Fundamental Parallelepipeds and Determinants

Let $B : \mathbb{R}^n \to \mathbb{R}^n$ be an invertible linear map. Define the lattice Λ and the fundamental parallelepiped $\mathcal{P}(B)$ by

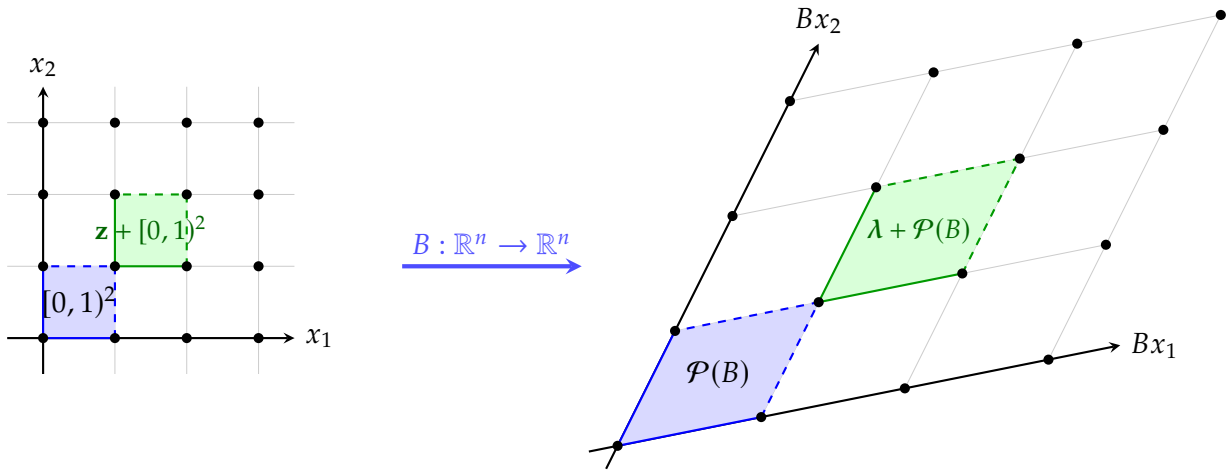
$$\Lambda = \mathcal{L}(B) = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\} \quad \text{and} \quad \mathcal{P}(B) = \{B\boldsymbol{\alpha} : \boldsymbol{\alpha} \in [0,1)^n\}.$$

Then $\mathbb{R}^n$ admits the following disjoint decomposition:

$$\mathbb{R}^n = \bigcup_{\lambda \in \Lambda} \big(\lambda + \mathcal{P}(B)\big),$$

where for any $\lambda_1, \lambda_2 \in \Lambda$,

$$\big(\lambda_1 + \mathcal{P}(B)\big) \cap \big(\lambda_2 + \mathcal{P}(B)\big) = \varnothing \quad \text{whenever} \quad \lambda_1 \neq \lambda_2.$$



21-24 Let

$$B = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \qquad \Lambda = \mathcal{L}(B), \qquad \mathcal{P}(B) = \{B\boldsymbol{\alpha} : \boldsymbol{\alpha} \in [0,1)^2\}.$$

21. Compute B^{-1}.

22. For each of the following vectors $\mathbf{x}$, compute

$$\mathbf{t} = B^{-1}\mathbf{x}, \qquad \mathbf{z} = \lfloor \mathbf{t} \rfloor \in \mathbb{Z}^2, \qquad \boldsymbol{\alpha} = \mathbf{t} - \lfloor \mathbf{t} \rfloor \in [0,1)^2,$$

and hence find the unique decomposition

$$\mathbf{x} = B\mathbf{z} + B\boldsymbol{\alpha}.$$

Use:

$$\begin{pmatrix} 7.4 \\ 5.2 \end{pmatrix}, \quad \begin{pmatrix} -1.3 \\ 4.7 \end{pmatrix}, \quad \begin{pmatrix} 3 \\ -2.4 \end{pmatrix}, \quad \begin{pmatrix} -5.8 \\ -0.2 \end{pmatrix} \quad \in \mathcal{L}(B).$$

23. For each vector above, identify explicitly the lattice vector

$$\mathbf{y} = B\mathbf{z} \in \mathcal{L}(B)$$

and the point

$$\mathbf{p} = B\boldsymbol{\alpha} \in \mathcal{P}(B)$$

such that $\mathbf{x} = \mathbf{y} + \mathbf{p}$.

24. Check in each case that $\mathbf{p} \in \mathcal{P}(B)$ by writing it in the form

$$\mathbf{p} = \alpha_1\mathbf{b}_1 + \alpha_2\mathbf{b}_2 \qquad \text{with } 0 \le \alpha_1, \alpha_2 < 1.$$

25-28 Let

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \qquad \Lambda = \mathcal{L}(B), \qquad \mathcal{P}(B) = \{B\boldsymbol{\alpha} : \boldsymbol{\alpha} \in [0,1)^2\}.$$

25. Compute the four vertices of $\mathcal{P}(B)$.

26. For each point $\mathbf{x}$ below, determine the unique lattice vector $\mathbf{y} \in \Lambda$ such that

$$\mathbf{x} \in \mathbf{y} + \mathcal{P}(B).$$

Also compute the corresponding $\mathbf{p} \in \mathcal{P}(B)$ with

$$\mathbf{x} = \mathbf{y} + \mathbf{p}.$$

Use:

$$\begin{pmatrix} 4.2 \\ 5.1 \end{pmatrix}, \qquad \begin{pmatrix} -0.7 \\ 1.6 \end{pmatrix}, \qquad \begin{pmatrix} 3.9 \\ -2.4 \end{pmatrix}, \qquad \begin{pmatrix} -3.3 \\ -4.8 \end{pmatrix} \qquad \in \mathcal{L}(B).$$

27. For each point, verify your answer by computing $B^{-1}\mathbf{x}$ and separating its coordinates into integer parts and fractional parts.

28. Pick one of the four points and draw the translate $\mathbf{y} + \mathcal{P}(B)$ that contains it.

29-33 Let

$$B = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix}, \qquad \Lambda = \mathcal{L}(B).$$

For $\mathbf{x} \in \mathbb{R}^3$, define its remainder modulo Λ by

$$r_B(\mathbf{x}) = B\big(\{B^{-1}\mathbf{x}\}\big),$$

where fractional parts are taken coordinatewise.

29. Compute $r_B(\mathbf{x})$ for each of the following vectors:

$$\mathbf{x}_1 = \begin{pmatrix} 5.2 \\ 3.7 \\ -1.4 \end{pmatrix}, \qquad \mathbf{x}_2 = \begin{pmatrix} -0.6 \\ 4.1 \\ 2.3 \end{pmatrix}, \qquad \mathbf{x}_3 = \begin{pmatrix} 7.2 \\ 5.7 \\ 0.6 \end{pmatrix}.$$

30. For each $\mathbf{x}_i$, also compute the corresponding lattice vector

$$\mathbf{y}_i = \mathbf{x}_i - r_B(\mathbf{x}_i) \in \Lambda.$$

31. Decide which pairs among $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ lie in the same coset modulo Λ. Justify your answer by comparing their remainders.

32. For every pair that lies in the same coset, compute explicitly the lattice vector

$$\mathbf{x}_i - \mathbf{x}_j \in \Lambda.$$

33. Explain how these calculations illustrate the uniqueness of the representation

$$\mathbf{x} = \mathbf{y} + \mathbf{p}, \qquad \mathbf{y} \in \Lambda,\ \mathbf{p} \in \mathcal{P}(B).$$

A.4 Successive Minima and Short Vector

A.5 Dual Lattices

A.6 Reduced Bases and the LLL algorithm

A.7 q-ary Lattices and SIS

A.8 Conceptual and Proof Exercises

Appendix B

Solutions for Lattice

B.1 Euclidean Spaces and Lattices

1. **Sol.**

$$\boxed{\langle \mathbf{x}, \mathbf{y} \rangle = 1, \qquad \|\mathbf{x}\| = \sqrt{14}, \qquad \|\mathbf{y}\| = \sqrt{17}, \qquad \|\mathbf{x} + \mathbf{y}\| = \sqrt{33}.}$$

□

2. **Sol.** (a) Yes, this is a lattice. Indeed,

$$\mathbb{Z}^2 = \{a(1,0) + b(0,1) : a, b \in \mathbb{Z}\} = \mathcal{L}(\mathbf{e}_1, \mathbf{e}_2).$$

(b) Yes, this is a lattice of rank 1 in $\mathbb{R}^2$. We may write it as

$$\{a(1, \sqrt{2}) : a \in \mathbb{Z}\} = \mathcal{L}(1, \sqrt{2}).$$

(c) Yes, this is a lattice in $\mathbb{R}^2$.

$$\{a(1,0) + b(1/2, 1) : a, b \in \mathbb{Z}\} = \mathcal{L}(\mathbf{b}_1, \mathbf{b}_2) \quad \text{where} \quad \mathbf{b}_1 = (1,0), \qquad \mathbf{b}_2 = \left(\frac{1}{2}, 1\right).$$

(d) No, this is not a lattice. There are rational points arbitrarily close to $(0,0)$, for example

$$\left(\frac{1}{n}, 0\right) \in \mathbb{Q}^2 \qquad \text{for all } n \in \mathbb{N},$$

and these converge to $(0,0)$. Hence $\mathbb{Q}^2$ is not a lattice.

Therefore:

$$\boxed{\mathbb{Z}^2,\ \{(a, \sqrt{2}a) : a \in \mathbb{Z}\},\ \{a(1,0) + b(1/2,1) : a, b \in \mathbb{Z}\} \text{ are lattices, but } \mathbb{Q}^2 \text{ is not.}}$$

□

3. **Sol.** The real span is

$$\operatorname{span}_{\mathbb{R}}\{\mathbf{b}_1, \mathbf{b}_2\} = \left\{ a \begin{pmatrix} 2 \\ 0 \end{pmatrix} + b \begin{pmatrix} 1 \\ 1 \end{pmatrix} : a, b \in \mathbb{R} \right\}.$$

Computing,

$$a \begin{pmatrix} 2 \\ 0 \end{pmatrix} + b \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2a + b \\ b \end{pmatrix}.$$

Given any $(x, y) \in \mathbb{R}^2$, choose $b = y$ and $a = (x - y)/2$. Hence every vector in $\mathbb{R}^2$ has this form, so

$$\operatorname{span}_{\mathbb{R}}\{\mathbf{b}_1, \mathbf{b}_2\} = \mathbb{R}^2.$$

The lattice generated by $\mathbf{b}_1, \mathbf{b}_2$ is

$$\Lambda = \left\{ m \begin{pmatrix} 2 \\ 0 \end{pmatrix} + n \begin{pmatrix} 1 \\ 1 \end{pmatrix} : m, n \in \mathbb{Z} \right\} = \left\{ \begin{pmatrix} 2m + n \\ n \end{pmatrix} : m, n \in \mathbb{Z} \right\}.$$

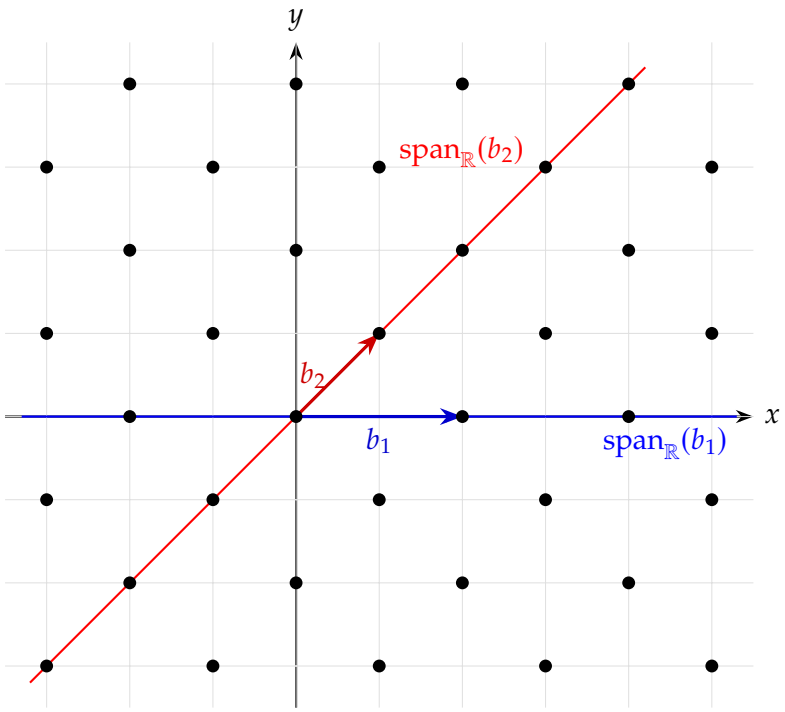
Equivalently,

$$\Lambda = \{(x, y) \in \mathbb{Z}^2 : x - y \in 2\mathbb{Z}\} = \{(x, y) \in \mathbb{Z}^2 : x \equiv y \pmod 2\}.$$

So the lattice consists of all integer points whose coordinates have the same parity. Therefore,

$$\boxed{\operatorname{span}_{\mathbb{R}}\{\mathbf{b}_1, \mathbf{b}_2\} = \mathbb{R}^2, \qquad \Lambda = \left\{ \begin{pmatrix} 2m + n \\ n \end{pmatrix} : m, n \in \mathbb{Z} \right\}.}$$

□



4. **Sol.** We first show that Λ is a subgroup of $(\mathbb{R}^n, +)$. If

$$\mathbf{x} = \sum_{i=1}^{m} z_i \mathbf{b}_i, \qquad \mathbf{y} = \sum_{i=1}^{m} w_i \mathbf{b}_i$$

are in Λ, with $z_i, w_i \in \mathbb{Z}$, then

$$\mathbf{x} - \mathbf{y} = \sum_{i=1}^{m} (z_i - w_i)\mathbf{b}_i.$$

Since $z_i - w_i \in \mathbb{Z}$, we have $\mathbf{x} - \mathbf{y} \in \Lambda$.

Let

$$V = \operatorname{span}_{\mathbb{R}}\{\mathbf{b}_1, \ldots, \mathbf{b}_m\}.$$

Since the vectors are linearly independent, the map

$$T : \mathbb{R}^m \to V, \qquad T(z_1, \ldots, z_m) = \sum_{i=1}^{m} z_i \mathbf{b}_i$$

is a linear isomorphism. In particular, $T(\mathbb{Z}^m) = \Lambda$. Because $\mathbb{Z}^m$ is discrete in $\mathbb{R}^m$, and T is a homeomorphism from $\mathbb{R}^m$ onto V, it follows that Λ is discrete in V, hence also discrete in $\mathbb{R}^n$. Therefore Λ is a discrete subgroup of $\mathbb{R}^n$. □

5. **Sol.** Consider the standard lattice

$$\mathbb{Z}^2.$$

One basis is

$$\mathbf{b}_1 = (1, 0), \qquad \mathbf{b}_2 = (0, 1).$$

Another basis is

$$\mathbf{c}_1 = (1, 1), \qquad \mathbf{c}_2 = (1, 0).$$

Indeed,

$$\mathbf{c}_1 = (1, 1), \qquad \mathbf{c}_2 = (1, 0)$$

are linearly independent, and every integer vector $(x, y) \in \mathbb{Z}^2$ can be written as

$$(x, y) = y(1, 1) + (x - y)(1, 0).$$

Since $x, y \in \mathbb{Z}$, this is an integer linear combination of $\mathbf{c}_1, \mathbf{c}_2$. Hence

$$\mathbb{Z}^2 = \mathcal{L}(\mathbf{b}_1, \mathbf{b}_2) = \mathcal{L}(\mathbf{c}_1, \mathbf{c}_2).$$

Thus two distinct bases of the same lattice are

$$\boxed{\{(1, 0), (0, 1)\} \qquad \text{and} \qquad \{(1, 1), (1, 0)\}.}$$

□

6. **Sol.** The lattice generated by $\mathbf{b}_1, \mathbf{b}_2$ is

$$\mathcal{L}(B) = \{z_1\mathbf{b}_1 + z_2\mathbf{b}_2 : z_1, z_2 \in \mathbb{Z}\}.$$

Substituting the basis vectors,

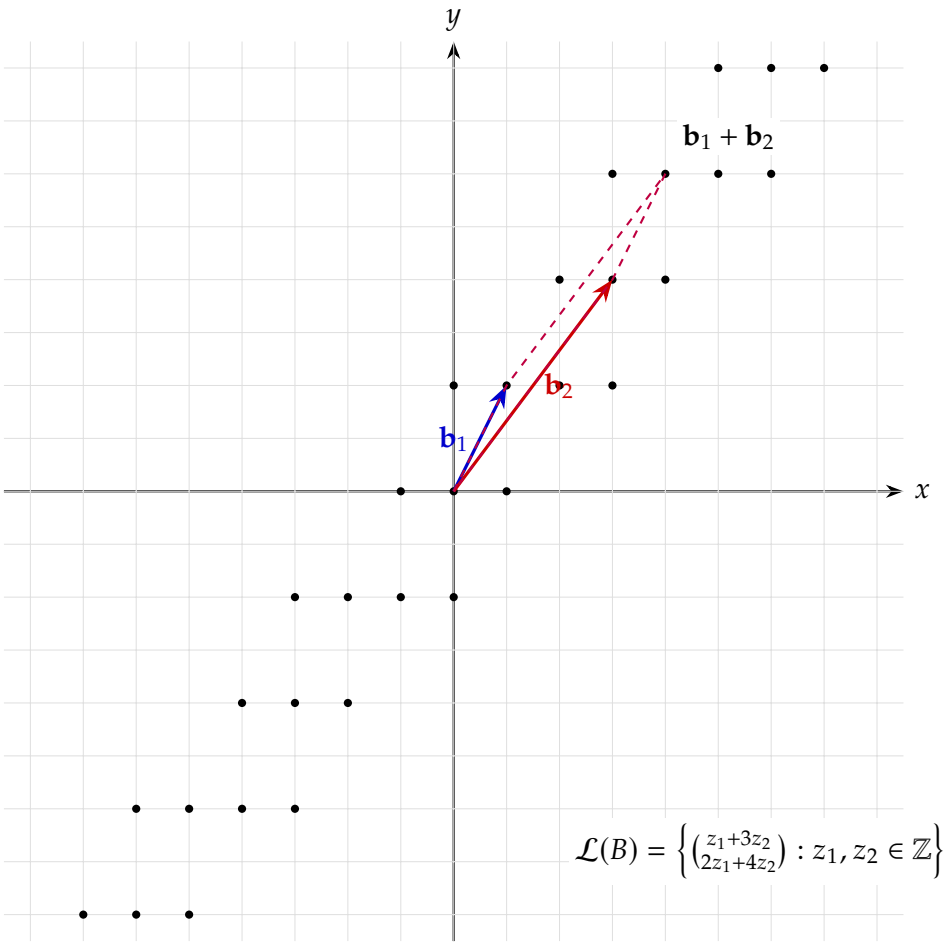
$$\mathcal{L}(B) = \left\{ z_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + z_2 \begin{pmatrix} 3 \\ 4 \end{pmatrix} : z_1, z_2 \in \mathbb{Z} \right\}.$$

Thus

$$\mathcal{L}(B) = \left\{ \begin{pmatrix} z_1 + 3z_2 \\ 2z_1 + 4z_2 \end{pmatrix} : z_1, z_2 \in \mathbb{Z} \right\}.$$

So explicitly,

$$\boxed{\mathcal{L}(B) = \left\{(x, y) \in \mathbb{R}^2 : x = z_1 + 3z_2,\ y = 2z_1 + 4z_2,\ z_1, z_2 \in \mathbb{Z}\right\}.}$$



□

7. **Sol.** Consider the lattice $\mathbb{Z}^2$. For each integer $k \in \mathbb{Z}$, define

$$\mathbf{b}_1^{(k)} = (1, 0), \qquad \mathbf{b}_2^{(k)} = (k, 1).$$

Then

$$B_k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$$

has $\det(B_k) = 1$, so its columns form a basis of $\mathbb{Z}^2$. Indeed, every vector in $\mathcal{L}(B_k)$ has the form

$$a(1, 0) + b(k, 1) = (a + bk, b), \qquad a, b \in \mathbb{Z}.$$

Given any $(x, y) \in \mathbb{Z}^2$, choose $b = y$ and $a = x - ky$. Then

$$(x, y) = a(1, 0) + b(k, 1),$$

so $\mathcal{L}(B_k) = \mathbb{Z}^2$. As k varies over $\mathbb{Z}$, these bases are distinct. Therefore the lattice $\mathbb{Z}^2$ has infinitely many bases. Hence a lattice may have infinitely many bases. □

8. **Sol.** Suppose

$$\Lambda = \left\{ \sum_{i=1}^{m} z_i \mathbf{b}_i : z_i \in \mathbb{Z} \right\}$$

for some linearly independent vectors $\mathbf{b}_1, \ldots, \mathbf{b}_m \in \mathbb{R}^n$. Define the map

$$f : \mathbb{Z}^m \to \Lambda, \qquad f(z_1, \ldots, z_m) = \sum_{i=1}^{m} z_i \mathbf{b}_i.$$

This map is surjective by definition of Λ. Since $\mathbb{Z}^m$ is countable, and the image of a countable set under a map is countable, it follows that Λ is countable. Therefore every lattice is countable. □

9. **Sol.** Let

$$\Lambda = \left\{ \sum_{i=1}^{m} z_i \mathbf{b}_i : z_i \in \mathbb{Z} \right\}.$$

First, Λ contains the zero vector because taking all coefficients equal to 0 gives

$$\mathbf{0} = \sum_{i=1}^{m} 0 \cdot \mathbf{b}_i \in \Lambda.$$

Now let

$$\mathbf{x} = \sum_{i=1}^{m} z_i \mathbf{b}_i, \qquad \mathbf{y} = \sum_{i=1}^{m} w_i \mathbf{b}_i$$

be vectors in Λ. Then

$$\mathbf{x} + \mathbf{y} = \sum_{i=1}^{m} (z_i + w_i) \mathbf{b}_i.$$

Since $z_i + w_i \in \mathbb{Z}$, we have $\mathbf{x} + \mathbf{y} \in \Lambda$. Thus Λ is closed under addition. Also,

$$-\mathbf{x} = \sum_{i=1}^{m} (-z_i) \mathbf{b}_i.$$

Since $-z_i \in \mathbb{Z}$, it follows that $-\mathbf{x} \in \Lambda$. Thus Λ is closed under negation. Therefore every lattice contains $\mathbf{0}$ and is closed under addition and negation. □

10. **Sol.** We rewrite the general element of Λ as

$$\Lambda = \{a(2,0) + b(1,1) : a, b \in \mathbb{Z}\}.$$

So a basis matrix is obtained by taking these vectors as columns:

$$B = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

Then

$$B \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2a + b \\ b \end{pmatrix}.$$

Therefore

$$\boxed{B = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}} \qquad \text{and} \qquad \Lambda = \mathcal{L}(B).$$

□

B.2 Gram–Schmidt Orthogonalization

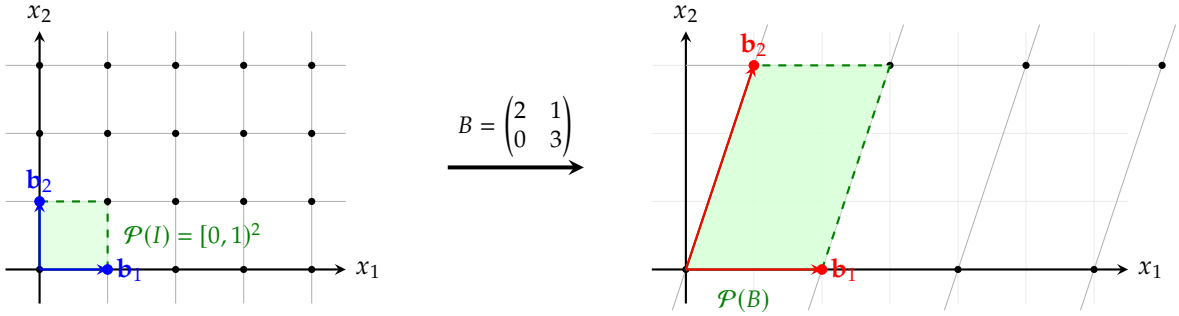
B.3 Fundamental Parallelepipeds and Determinants

21-24 We are given

$$B = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \qquad \Lambda = \mathcal{L}(B), \qquad \mathcal{P}(B) = \{B\boldsymbol{\alpha} : \boldsymbol{\alpha} \in [0,1)^2\}.$$

The columns of B are

$$\mathbf{b}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \qquad \mathbf{b}_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \qquad \text{so} \qquad B = \begin{pmatrix} \mathbf{b}_1 & \mathbf{b}_2 \end{pmatrix}.$$



21. **Sol.** Since

$$\det(B) = 2 \cdot 3 - 0 \cdot 1 = 6,$$

we have

$$B^{-1} = \frac{1}{6}\begin{pmatrix} 3 & -1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{6} \\ 0 & \frac{1}{3} \end{pmatrix}.$$

Thus

$$\boxed{B^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{6} \\ 0 & \frac{1}{3} \end{pmatrix}.}$$

□

22. **Sol.** (a) Write $\mathbf{x} = \begin{pmatrix} 7.4 \\ 5.2 \end{pmatrix} = \begin{pmatrix} \frac{37}{5} \\ \frac{26}{5} \end{pmatrix}$. Then

$$\mathbf{t} = B^{-1}\mathbf{x} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{6} \\ 0 & \frac{1}{3} \end{pmatrix}\begin{pmatrix} \frac{37}{5} \\ \frac{26}{5} \end{pmatrix} = \begin{pmatrix} \frac{17}{6} \\ \frac{26}{15} \end{pmatrix}.$$

Hence

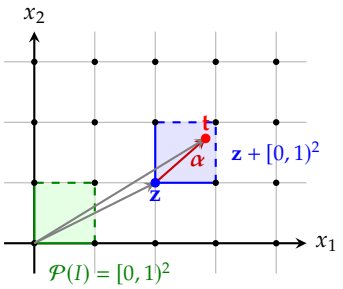
$$\mathbf{z} = \lfloor \mathbf{t} \rfloor = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \mathbf{t} - \mathbf{z} = \begin{pmatrix} \frac{5}{6} \\ \frac{11}{15} \end{pmatrix}.$$

Therefore

$$B\mathbf{z} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}, \qquad \text{and} \qquad B\boldsymbol{\alpha} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} \frac{5}{6} \\ \frac{11}{15} \end{pmatrix} = \begin{pmatrix} \frac{12}{5} \\ \frac{11}{5} \end{pmatrix}.$$

So

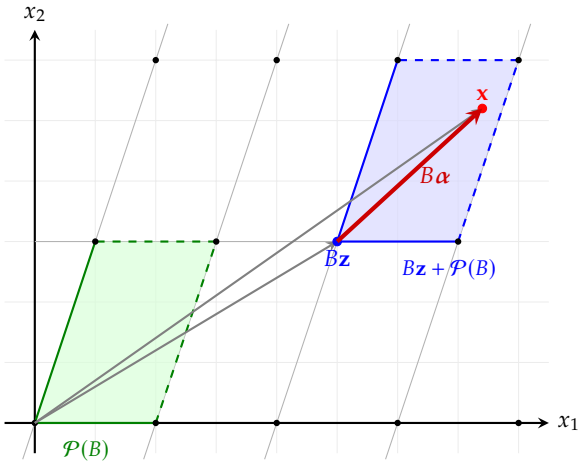
$$\boxed{\begin{pmatrix} 7.4 \\ 5.2 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix} + \begin{pmatrix} \frac{12}{5} \\ \frac{11}{5} \end{pmatrix}.}$$



$$\mathcal{L}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$$

$$B=\begin{pmatrix}2 & 1\\ 0 & 3\end{pmatrix}$$

$$B^{-1}=\begin{pmatrix}\frac{1}{2} & -\frac{1}{6}\\ 0 & \frac{1}{3}\end{pmatrix}$$



$$\mathcal{L}\begin{pmatrix}2 & 1\\ 0 & 3\end{pmatrix} = \left\{\begin{pmatrix}2 & 1\\ 0 & 3\end{pmatrix}\mathbf{z} : \mathbf{z}\in[0,1)^n\right\}$$

(b) Write $\mathbf{x}=\begin{pmatrix}-\frac{13}{10}\\ \frac{47}{10}\end{pmatrix}$. Then

$$\mathbf{t}=B^{-1}\mathbf{x}=\begin{pmatrix}\frac{1}{2} & -\frac{1}{6}\\ 0 & \frac{1}{3}\end{pmatrix}\begin{pmatrix}-\frac{13}{10}\\ \frac{47}{10}\end{pmatrix}=\begin{pmatrix}-\frac{43}{30}\\ \frac{47}{30}\end{pmatrix}.$$

Hence

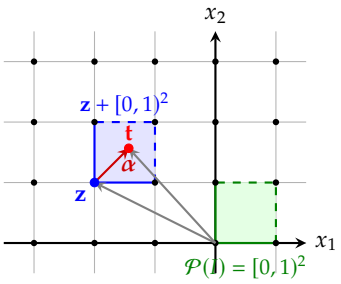
$$\mathbf{z}=\lfloor\mathbf{t}\rfloor=\begin{pmatrix}-2\\ 1\end{pmatrix},\qquad \boldsymbol{\alpha}=\mathbf{t}-\mathbf{z}=\begin{pmatrix}\frac{17}{30}\\ \frac{17}{30}\end{pmatrix}.$$

Therefore

$$B\mathbf{z}=\begin{pmatrix}2 & 1\\ 0 & 3\end{pmatrix}\begin{pmatrix}-2\\ 1\end{pmatrix}=\begin{pmatrix}-3\\ 3\end{pmatrix},\quad\text{and}\quad B\boldsymbol{\alpha}=\begin{pmatrix}2 & 1\\ 0 & 3\end{pmatrix}\begin{pmatrix}\frac{17}{30}\\ \frac{17}{30}\end{pmatrix}=\begin{pmatrix}\frac{17}{10}\\ \frac{17}{10}\end{pmatrix}.$$

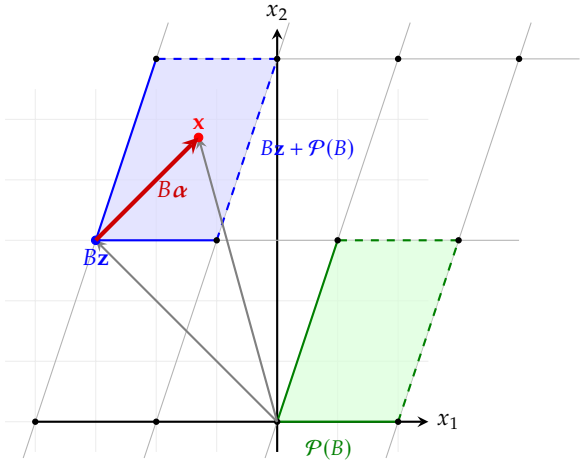
So

$$\boxed{\begin{pmatrix}-1.3\\ 4.7\end{pmatrix}=\begin{pmatrix}-3\\ 3\end{pmatrix}+\begin{pmatrix}\frac{17}{10}\\ \frac{17}{10}\end{pmatrix}.}$$



$$B=\begin{pmatrix}2 & 1\\ 0 & 3\end{pmatrix}$$

$$B^{-1}=\begin{pmatrix}\frac{1}{2} & -\frac{1}{6}\\ 0 & \frac{1}{3}\end{pmatrix}$$



(c) Write $\mathbf{x} = \begin{pmatrix} 3 \\ -\frac{12}{5} \end{pmatrix}$. Then

$$\mathbf{t} = B^{-1}\mathbf{x} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{6} \\ 0 & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 3 \\ -\frac{12}{5} \end{pmatrix} = \begin{pmatrix} \frac{19}{10} \\ -\frac{4}{5} \end{pmatrix}.$$

Hence

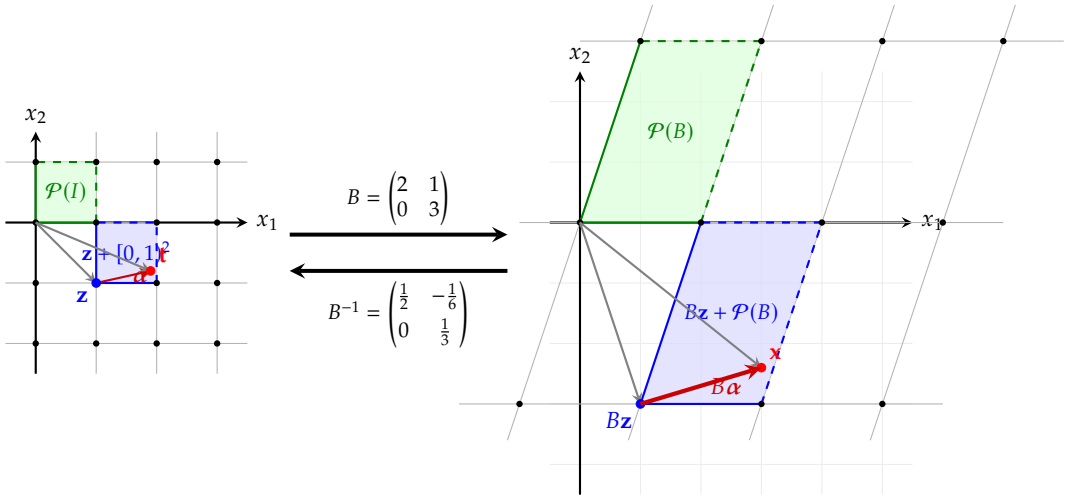
$$\mathbf{z} = \lfloor \mathbf{t} \rfloor = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \mathbf{t} - \mathbf{z} = \begin{pmatrix} \frac{9}{10} \\ \frac{1}{5} \end{pmatrix}.$$

Therefore

$$B\mathbf{z} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \end{pmatrix}, \quad \text{and} \quad B\boldsymbol{\alpha} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} \frac{9}{10} \\ \frac{1}{5} \end{pmatrix} = \begin{pmatrix} 2 \\ \frac{3}{5} \end{pmatrix}.$$

So

$$\boxed{\begin{pmatrix} 3 \\ -2.4 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \end{pmatrix} + \begin{pmatrix} 2 \\ \frac{3}{5} \end{pmatrix}.}$$



(d) Write $\mathbf{x} = \begin{pmatrix} -\frac{29}{5} \\ -\frac{1}{5} \end{pmatrix}$. Then

$$\mathbf{t} = B^{-1}\mathbf{x} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{6} \\ 0 & \frac{1}{3} \end{pmatrix} \begin{pmatrix} -\frac{29}{5} \\ -\frac{1}{5} \end{pmatrix} = \begin{pmatrix} -\frac{43}{15} \\ -\frac{1}{15} \end{pmatrix}.$$

Hence

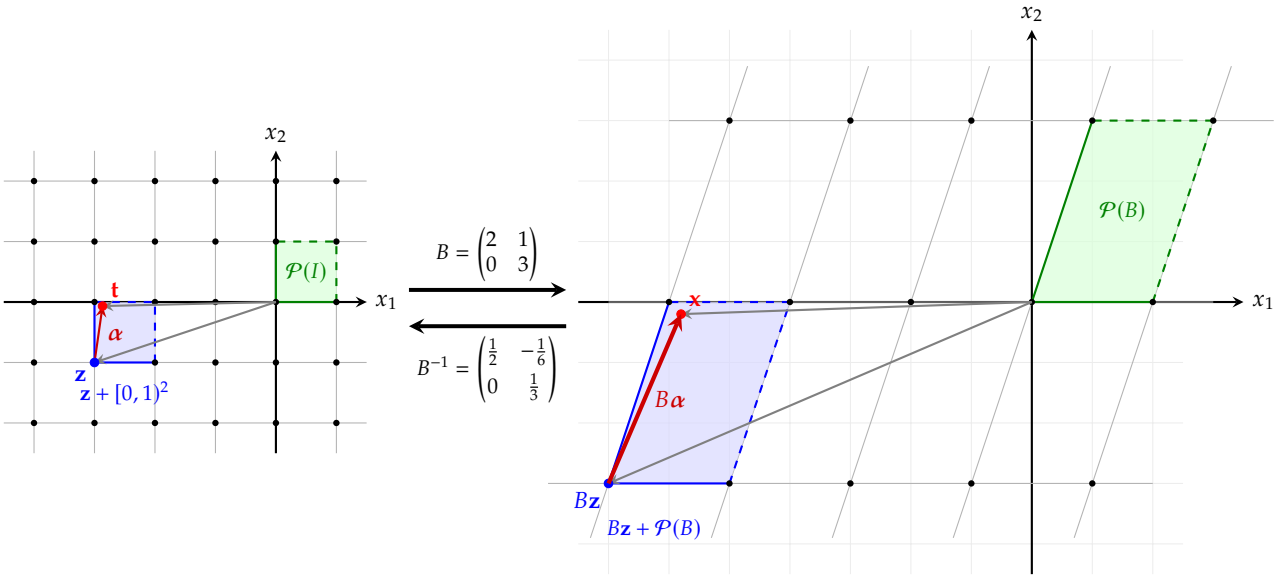
$$\mathbf{z} = \lfloor \mathbf{t} \rfloor = \begin{pmatrix} -3 \\ -1 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \mathbf{t} - \mathbf{z} = \begin{pmatrix} \frac{2}{15} \\ \frac{14}{15} \end{pmatrix}.$$

Therefore

$$B\mathbf{z} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} -3 \\ -1 \end{pmatrix} = \begin{pmatrix} -7 \\ -3 \end{pmatrix}, \quad \text{and} \quad B\boldsymbol{\alpha} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} \frac{2}{15} \\ \frac{14}{15} \end{pmatrix} = \begin{pmatrix} \frac{6}{5} \\ \frac{14}{5} \end{pmatrix}.$$

So

$$\boxed{\begin{pmatrix} -5.8 \\ -0.2 \end{pmatrix} = \begin{pmatrix} -7 \\ -3 \end{pmatrix} + \begin{pmatrix} \frac{6}{5} \\ \frac{14}{5} \end{pmatrix}.}$$



□

23. **Sol.**

$$\boxed{\begin{pmatrix}7.4\\5.2\end{pmatrix}=\begin{pmatrix}5\\3\end{pmatrix}+\begin{pmatrix}\frac{12}{5}\\\frac{11}{5}\end{pmatrix},\qquad \begin{pmatrix}-1.3\\4.7\end{pmatrix}=\begin{pmatrix}-3\\3\end{pmatrix}+\begin{pmatrix}\frac{17}{10}\\\frac{17}{10}\end{pmatrix},}$$
$$\boxed{\begin{pmatrix}3\\-2.4\end{pmatrix}=\begin{pmatrix}1\\-3\end{pmatrix}+\begin{pmatrix}2\\\frac{3}{5}\end{pmatrix},\qquad \begin{pmatrix}-5.8\\-0.2\end{pmatrix}=\begin{pmatrix}-7\\-3\end{pmatrix}+\begin{pmatrix}\frac{6}{5}\\\frac{14}{5}\end{pmatrix}.}$$

□

24. **Sol.** Recall that

$$\mathbf{b}_1=\begin{pmatrix}2\\0\end{pmatrix},\qquad \mathbf{b}_2=\begin{pmatrix}1\\3\end{pmatrix}.$$

$$\mathbf{p}=\begin{pmatrix}\frac{12}{5}\\\frac{11}{5}\end{pmatrix}=\frac{5}{6}\mathbf{b}_1+\frac{11}{15}\mathbf{b}_2\implies 0\le\frac{5}{6}<1,\qquad 0\le\frac{11}{15}<1.$$
$$\mathbf{p}=\begin{pmatrix}\frac{17}{10}\\\frac{17}{10}\end{pmatrix}=\frac{17}{30}\mathbf{b}_1+\frac{17}{30}\mathbf{b}_2\implies 0\le\frac{17}{30}<1.$$
$$\mathbf{p}=\begin{pmatrix}2\\\frac{3}{5}\end{pmatrix}=\frac{9}{10}\mathbf{b}_1+\frac{1}{5}\mathbf{b}_2\implies 0\le\frac{9}{10}<1,\qquad 0\le\frac{1}{5}<1.$$
$$\mathbf{p}=\begin{pmatrix}\frac{6}{5}\\\frac{14}{5}\end{pmatrix}=\frac{2}{15}\mathbf{b}_1+\frac{14}{15}\mathbf{b}_2,\implies 0\le\frac{2}{15}<1,\qquad 0\le\frac{14}{15}<1.$$

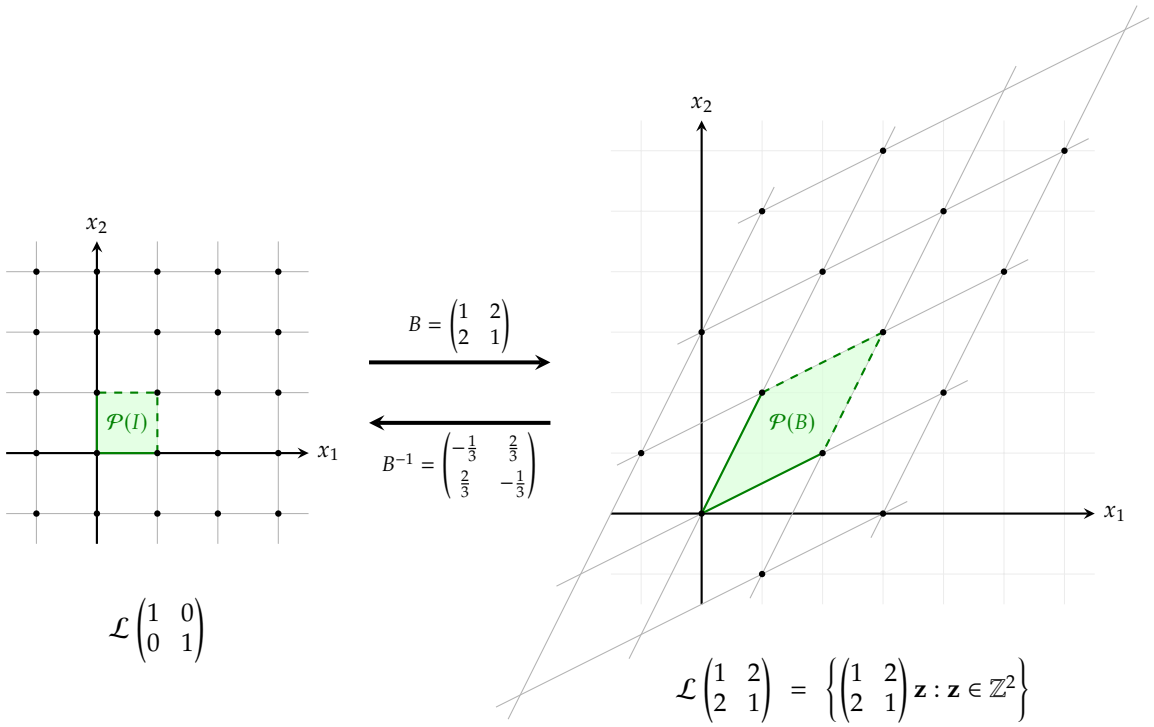
□

25-28 We are given

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \qquad \Lambda = \mathcal{L}(B), \qquad \mathcal{P}(B) = \{B\boldsymbol{\alpha} : \boldsymbol{\alpha} \in [0,1)^2\}.$$

The columns of B are

$$\mathbf{b}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad \mathbf{b}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$



25. **Sol.** Since

$$\mathcal{P}(B) = \{\alpha_1\mathbf{b}_1 + \alpha_2\mathbf{b}_2 : 0 \le \alpha_1, \alpha_2 < 1\},$$

the associated parallelogram has vertices

$$\mathbf{0}, \qquad \mathbf{b}_1, \qquad \mathbf{b}_2, \qquad \mathbf{b}_1 + \mathbf{b}_2.$$

Thus the four vertices are

$$\boxed{\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \qquad \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad \begin{pmatrix} 3 \\ 3 \end{pmatrix}.}$$

Remark. Because $\mathcal{P}(B)$ is defined using $[0,1)^2$, it is a half-open parallelogram. Geometrically, however, these are the four corner points of its closure. □

26. **Sol.** We compute

$$\det(B) = 1 \cdot 1 - 2 \cdot 2 = -3,$$

so

$$B^{-1} = \frac{1}{-3}\begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{pmatrix}.$$

(a) Write $\mathbf{x} = \begin{pmatrix} \frac{21}{5} \\ \frac{51}{10} \end{pmatrix}$. Then

$$B^{-1}\mathbf{x} = \begin{pmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{pmatrix}\begin{pmatrix} \frac{21}{5} \\ \frac{51}{10} \end{pmatrix} = \begin{pmatrix} 2 \\ \frac{11}{10} \end{pmatrix}.$$

Hence

$$\mathbf{z} = \lfloor B^{-1}\mathbf{x} \rfloor = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \begin{pmatrix} 0 \\ \frac{1}{10} \end{pmatrix}.$$

Therefore

$$\mathbf{y} = B\mathbf{z} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix} \quad \text{and} \quad \mathbf{p} = B\boldsymbol{\alpha} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ \frac{1}{10} \end{pmatrix} = \begin{pmatrix} \frac{1}{5} \\ \frac{1}{10} \end{pmatrix}.$$

So

$$\boxed{\begin{pmatrix} 4.2 \\ 5.1 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix} + \begin{pmatrix} \frac{1}{5} \\ \frac{1}{10} \end{pmatrix}.}$$

(b) Write$\mathbf{x} = \begin{pmatrix} -\frac{7}{10} \\ \frac{8}{5} \end{pmatrix}$. Then

$$B^{-1}\mathbf{x} = \begin{pmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{pmatrix}\begin{pmatrix} -\frac{7}{10} \\ \frac{8}{5} \end{pmatrix} = \begin{pmatrix} \frac{13}{10} \\ -1 \end{pmatrix}.$$

Hence

$$\mathbf{z} = \lfloor B^{-1}\mathbf{x} \rfloor = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \begin{pmatrix} \frac{3}{10} \\ 0 \end{pmatrix}.$$

Therefore

$$\mathbf{y} = B\mathbf{z} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad \text{and} \quad \mathbf{p} = B\boldsymbol{\alpha} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} \frac{3}{10} \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{3}{10} \\ \frac{3}{5} \end{pmatrix}.$$

So

$$\boxed{\begin{pmatrix} -0.7 \\ 1.6 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} \frac{3}{10} \\ \frac{3}{5} \end{pmatrix}.}$$

(c) Write $\mathbf{x} = \begin{pmatrix} \frac{39}{10} \\ -\frac{12}{5} \end{pmatrix}$. Then

$$B^{-1}\mathbf{x} = \begin{pmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{pmatrix} \begin{pmatrix} \frac{39}{10} \\ -\frac{12}{5} \end{pmatrix} = \begin{pmatrix} -\frac{29}{10} \\ \frac{17}{5} \end{pmatrix}.$$

Hence

$$\mathbf{z} = \lfloor B^{-1}\mathbf{x} \rfloor = \begin{pmatrix} -3 \\ 3 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \begin{pmatrix} \frac{1}{10} \\ \frac{2}{5} \end{pmatrix}.$$

Therefore

$$\mathbf{y} = B\mathbf{z} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} -3 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \end{pmatrix}, \quad \text{and} \quad \mathbf{p} = B\boldsymbol{\alpha} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{10} \\ \frac{2}{5} \end{pmatrix} = \begin{pmatrix} \frac{9}{10} \\ \frac{3}{5} \end{pmatrix}.$$

So

$$\boxed{\begin{pmatrix} 3.9 \\ -2.4 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \end{pmatrix} + \begin{pmatrix} \frac{9}{10} \\ \frac{3}{5} \end{pmatrix}.}$$

(d) Write $\mathbf{x} = \begin{pmatrix} -\frac{33}{10} \\ -\frac{24}{5} \end{pmatrix}$. Then

$$B^{-1}\mathbf{x} = \begin{pmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{pmatrix} \begin{pmatrix} -\frac{33}{10} \\ -\frac{24}{5} \end{pmatrix} = \begin{pmatrix} -\frac{21}{10} \\ -\frac{3}{5} \end{pmatrix}.$$

Hence

$$\mathbf{z} = \lfloor B^{-1}\mathbf{x} \rfloor = \begin{pmatrix} -3 \\ -1 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \begin{pmatrix} \frac{9}{10} \\ \frac{2}{5} \end{pmatrix}.$$

Therefore

$$\mathbf{y} = B\mathbf{z} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} -3 \\ -1 \end{pmatrix} = \begin{pmatrix} -5 \\ -7 \end{pmatrix}, \quad \text{and} \quad \mathbf{p} = B\boldsymbol{\alpha} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} \frac{9}{10} \\ \frac{2}{5} \end{pmatrix} = \begin{pmatrix} \frac{17}{10} \\ \frac{11}{5} \end{pmatrix}.$$

So

$$\boxed{\begin{pmatrix} -3.3 \\ -4.8 \end{pmatrix} = \begin{pmatrix} -5 \\ -7 \end{pmatrix} + \begin{pmatrix} \frac{17}{10} \\ \frac{11}{5} \end{pmatrix}.}$$

□

27. **Sol.**

$$\begin{pmatrix} 4.2 \\ 5.1 \end{pmatrix}: \qquad B^{-1}\mathbf{x} = \begin{pmatrix} 2 \\ \frac{11}{10} \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{10} \end{pmatrix}.$$

$$\begin{pmatrix} -0.7 \\ 1.6 \end{pmatrix}: \qquad B^{-1}\mathbf{x} = \begin{pmatrix} \frac{13}{10} \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \begin{pmatrix} \frac{3}{10} \\ 0 \end{pmatrix}.$$

$$\begin{pmatrix} 3.9 \\ -2.4 \end{pmatrix}: \qquad B^{-1}\mathbf{x} = \begin{pmatrix} -\frac{29}{10} \\ \frac{17}{5} \end{pmatrix} = \begin{pmatrix} -3 \\ 3 \end{pmatrix} + \begin{pmatrix} \frac{1}{10} \\ \frac{2}{5} \end{pmatrix}.$$

$$\begin{pmatrix} -3.3 \\ -4.8 \end{pmatrix}: \qquad B^{-1}\mathbf{x} = \begin{pmatrix} -\frac{21}{10} \\ -\frac{3}{5} \end{pmatrix} = \begin{pmatrix} -3 \\ -1 \end{pmatrix} + \begin{pmatrix} \frac{9}{10} \\ \frac{2}{5} \end{pmatrix}.$$

□

28. **Sol.** Take $\mathbf{x} = \begin{pmatrix} 4.2 \\ 5.1 \end{pmatrix}$. From part (26), the corresponding lattice vector is

$$\mathbf{y} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}.$$

Therefore the translate containing $\mathbf{x}$ is

$$\mathbf{y} + \mathcal{P}(B) = \begin{pmatrix} 4 \\ 5 \end{pmatrix} + \mathcal{P}(B).$$

Its four vertices are

$$\begin{pmatrix} 4 \\ 5 \end{pmatrix}, \quad \begin{pmatrix} 4 \\ 5 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \end{pmatrix}, \quad \begin{pmatrix} 4 \\ 5 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 6 \end{pmatrix}, \quad \begin{pmatrix} 4 \\ 5 \end{pmatrix} + \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 7 \\ 8 \end{pmatrix}.$$

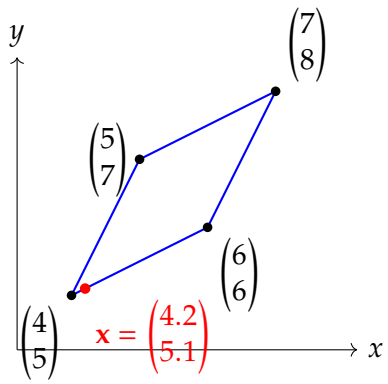
So one should draw the parallelogram with vertices

$$\boxed{\begin{pmatrix} 4 \\ 5 \end{pmatrix}, \quad \begin{pmatrix} 5 \\ 7 \end{pmatrix}, \quad \begin{pmatrix} 7 \\ 8 \end{pmatrix}, \quad \begin{pmatrix} 6 \\ 6 \end{pmatrix}}$$

and mark the point

$$\mathbf{x} = \begin{pmatrix} 4.2 \\ 5.1 \end{pmatrix}$$

inside it.



□

29-33 We are given

$$B = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix}, \qquad \Lambda = \mathcal{L}(B), \qquad r_B(\mathbf{x}) = B\big(\{B^{-1}\mathbf{x}\}\big).$$

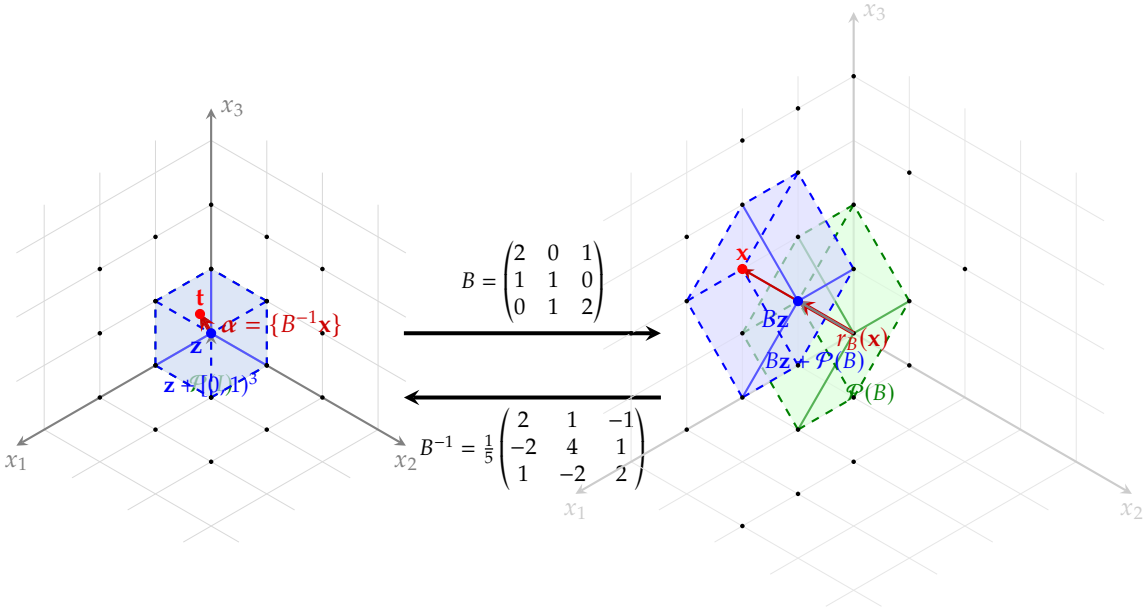
Compute B^{-1}: Since $\det(B) = 5$, the matrix B is invertible, and

$$B^{-1} = \frac{1}{5}\begin{pmatrix} 2 & 1 & -1 \\ -2 & 4 & 1 \\ 1 & -2 & 2 \end{pmatrix} = \begin{pmatrix} \frac{2}{5} & \frac{1}{5} & -\frac{1}{5} \\ -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \\ \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \end{pmatrix}.$$

Recall that for a real number t,

$$\{t\} = t - \lfloor t \rfloor \in [0, 1),$$

and fractional parts are taken coordinatewise.



$\mathcal{L}(I)$ $\qquad\qquad$ $\Lambda \;=\; \mathcal{L}(B)$

29. **Sol.** ($\mathbf{x}_1$) Note that

$$B^{-1}\mathbf{x}_1 = \begin{pmatrix} \frac{2}{5} & \frac{1}{5} & -\frac{1}{5} \\ -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \\ \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \end{pmatrix}\begin{pmatrix} \frac{26}{5} \\ \frac{37}{10} \\ -\frac{7}{5} \end{pmatrix} = \begin{pmatrix} \frac{31}{10} \\ \frac{3}{5} \\ -1 \end{pmatrix} \Longrightarrow \{B^{-1}\mathbf{x}_1\} = \begin{pmatrix} \{\frac{31}{10}\} \\ \{\frac{3}{5}\} \\ \{-1\} \end{pmatrix} = \begin{pmatrix} \frac{1}{10} \\ \frac{3}{5} \\ 0 \end{pmatrix}.$$

Therefore

$$r_B(\mathbf{x}_1) = B\begin{pmatrix} \frac{1}{10} \\ \frac{3}{5} \\ 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix}\begin{pmatrix} \frac{1}{10} \\ \frac{3}{5} \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{5} \\ \frac{7}{10} \\ \frac{3}{5} \end{pmatrix}.$$

($\mathbf{x}_2$) Note that

$$B^{-1}\mathbf{x}_2 = \begin{pmatrix} \frac{2}{5} & \frac{1}{5} & -\frac{1}{5} \\ -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \\ \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \end{pmatrix} \begin{pmatrix} -\frac{3}{5} \\ \frac{41}{10} \\ \frac{23}{10} \end{pmatrix} = \begin{pmatrix} \frac{3}{25} \\ \frac{199}{50} \\ -\frac{21}{25} \end{pmatrix} \Longrightarrow \left\{B^{-1}\mathbf{x}_2\right\} = \begin{pmatrix} \frac{3}{25} \\ \frac{49}{50} \\ \frac{4}{25} \end{pmatrix}.$$

Therefore

$$r_B(\mathbf{x}_2) = B \begin{pmatrix} \frac{3}{25} \\ \frac{49}{50} \\ \frac{4}{25} \end{pmatrix} = \begin{pmatrix} \frac{2}{5} \\ \frac{11}{10} \\ \frac{13}{10} \end{pmatrix}.$$

($\mathbf{x}_3$) Note that

$$B^{-1}\mathbf{x}_3 = \begin{pmatrix} \frac{2}{5} & \frac{1}{5} & -\frac{1}{5} \\ -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \\ \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \end{pmatrix} \begin{pmatrix} \frac{36}{5} \\ \frac{57}{10} \\ \frac{3}{5} \end{pmatrix} = \begin{pmatrix} \frac{39}{10} \\ \frac{9}{5} \\ -\frac{3}{5} \end{pmatrix} \Longrightarrow \left\{B^{-1}\mathbf{x}_3\right\} = \begin{pmatrix} \frac{9}{10} \\ \frac{4}{5} \\ \frac{2}{5} \end{pmatrix}.$$

Therefore

$$r_B(\mathbf{x}_3) = B \begin{pmatrix} \frac{9}{10} \\ \frac{4}{5} \\ \frac{2}{5} \end{pmatrix} = \begin{pmatrix} \frac{11}{5} \\ \frac{17}{10} \\ \frac{8}{5} \end{pmatrix}.$$

So the remainders are

$$\boxed{r_B(\mathbf{x}_1) = \begin{pmatrix} \frac{1}{5} \\ \frac{7}{10} \\ \frac{3}{5} \end{pmatrix}, \qquad r_B(\mathbf{x}_2) = \begin{pmatrix} \frac{2}{5} \\ \frac{11}{10} \\ \frac{13}{10} \end{pmatrix}, \qquad r_B(\mathbf{x}_3) = \begin{pmatrix} \frac{11}{5} \\ \frac{17}{10} \\ \frac{8}{5} \end{pmatrix}.}$$

□

30. **Sol.** ($\mathbf{x}_1$)

$$\mathbf{y}_1 = \mathbf{x}_1 - r_B(\mathbf{x}_1) = \begin{pmatrix} \frac{26}{5} \\ \frac{37}{10} \\ -\frac{7}{5} \end{pmatrix} - \begin{pmatrix} \frac{1}{5} \\ \frac{7}{10} \\ \frac{3}{5} \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \\ -2 \end{pmatrix}.$$

Also,

$$\mathbf{y}_1 = B \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix},$$

so $\mathbf{y}_1 \in \Lambda$.

($\mathbf{x}_2$)

$$\mathbf{y}_2 = \mathbf{x}_2 - r_B(\mathbf{x}_2) = \begin{pmatrix} -\frac{3}{5} \\ \frac{41}{10} \\ \frac{23}{10} \end{pmatrix} - \begin{pmatrix} \frac{2}{5} \\ \frac{11}{10} \\ \frac{13}{10} \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}.$$

Also,

$$\mathbf{y}_2 = B \begin{pmatrix} 0 \\ 3 \\ -1 \end{pmatrix},$$

so $\mathbf{y}_2 \in \Lambda$.

$(\mathbf{x}_3)$

$$\mathbf{y}_3 = \mathbf{x}_3 - r_B(\mathbf{x}_3) = \begin{pmatrix} \frac{36}{5} \\ \frac{57}{10} \\ \frac{3}{5} \end{pmatrix} - \begin{pmatrix} \frac{11}{5} \\ \frac{17}{10} \\ \frac{8}{5} \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ -1 \end{pmatrix}.$$

Also,

$$\mathbf{y}_3 = B \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix},$$

so $\mathbf{y}_3 \in \Lambda$.

Thus

$$\boxed{\mathbf{y}_1 = \begin{pmatrix} 5 \\ 3 \\ -2 \end{pmatrix}, \qquad \mathbf{y}_2 = \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}, \qquad \mathbf{y}_3 = \begin{pmatrix} 5 \\ 4 \\ -1 \end{pmatrix}.}$$

□

31. **Sol.** Two vectors lie in the same coset modulo Λ if and only if they have the same remainder modulo Λ. From part (29),

$$r_B(\mathbf{x}_1) \neq r_B(\mathbf{x}_2), \qquad r_B(\mathbf{x}_1) \neq r_B(\mathbf{x}_3), \qquad r_B(\mathbf{x}_2) \neq r_B(\mathbf{x}_3).$$

Therefore no two of the vectors lie in the same coset modulo Λ. Hence

$$\boxed{\text{none of the pairs } (\mathbf{x}_1, \mathbf{x}_2),\ (\mathbf{x}_1, \mathbf{x}_3),\ (\mathbf{x}_2, \mathbf{x}_3) \text{ lie in the same coset mod } \Lambda.}$$

□

32. **Sol.** Since no pair lies in the same coset modulo Λ, there are no such pairs. Hence there is no pair (i, j) with

$$\mathbf{x}_i - \mathbf{x}_j \in \Lambda.$$

Equivalently, one may check directly that none of the pairwise differences has integer B-coordinates:

$$B^{-1}(\mathbf{x}_1 - \mathbf{x}_2) = \begin{pmatrix} \frac{149}{50} \\ -\frac{169}{50} \\ -\frac{4}{25} \end{pmatrix}, \quad B^{-1}(\mathbf{x}_1 - \mathbf{x}_3) = \begin{pmatrix} -\frac{4}{5} \\ -\frac{6}{5} \\ -\frac{2}{5} \end{pmatrix}, \quad B^{-1}(\mathbf{x}_2 - \mathbf{x}_3) = \begin{pmatrix} -\frac{189}{50} \\ \frac{109}{50} \\ -\frac{6}{25} \end{pmatrix},$$

and none of these vectors lies in $\mathbb{Z}^3$. Therefore none of the differences belongs to Λ. □

33. **Sol.** For each $\mathbf{x}_i$, we computed

$$\mathbf{x}_i = \mathbf{y}_i + r_B(\mathbf{x}_i), \qquad \mathbf{y}_i \in \Lambda.$$

Moreover, since

$$\{B^{-1}\mathbf{x}_i\} \in [0, 1)^3,$$

each remainder

$$r_B(\mathbf{x}_i) = B\big(\{B^{-1}\mathbf{x}_i\}\big)$$

lies in the fundamental parallelepiped

$$\mathcal{P}(B) = \{B\boldsymbol{\alpha} : \boldsymbol{\alpha} \in [0,1)^3\}.$$

Thus each $\mathbf{x}_i$ has been written in the form

$$\mathbf{x}_i = \mathbf{y}_i + \mathbf{p}_i, \qquad \mathbf{y}_i \in \Lambda,\ \mathbf{p}_i \in \mathcal{P}(B).$$

The fact that the three remainders are different shows that the three vectors lie in different cosets modulo Λ. In other words, the remainder $\mathbf{p} \in \mathcal{P}(B)$ serves as the unique representative of the coset $\mathbf{x} + \Lambda$.

So these calculations concretely illustrate the theorem:

> Every $\mathbf{x} \in \mathbb{R}^3$ can be written uniquely as
>
> $$\mathbf{x} = \mathbf{y} + \mathbf{p}, \qquad \mathbf{y} \in \Lambda,\ \mathbf{p} \in \mathcal{P}(B).$$

Here the lattice part is $\mathbf{y}_i$, and the unique fundamental-domain representative is $r_B(\mathbf{x}_i)$. □